

3D Design (CAD) Pre-Study — 10 Hour Plan

(Fusion 360, mapped to internship Table 1 — 3D 설계(CAD) row)

Internship table field	Value
Area	3D 설계 (CAD)
Class hours	10h
On-site work	Part/housing modeling, assembly & tolerance design
Matching subject	컴퓨터그래픽스 / CAD 기초 (비교과)
Grade	2-3rd year
Content covered	3D 형상 모델링 (3D shape modeling), 파라메트릭 설계 (parametric design)

Everything below is scoped to those two curriculum items and the two on-site tasks, in **10 hours total**, split into 5 sessions of 2 hours, using **Fusion 360** (free for students/personal use).

Coverage map

교과 내용	Covered in
3D 형상 모델링 (basic shapes → solids)	Session 1, 2
부품 모델링 (on-site task: part modeling)	Session 2
조립 (on-site task: assembly)	Session 3
파라메트릭 설계	Session 4
치수 설계 (on-site task: tolerancing)	Session 4
하우징/최수 설계 → export for print	Session 5

Folder structure

```
cad-design-prep/
├── README.md
├── session-01-sketch-fundamentals/      <- 2D sketch → constraints (2h)
│   └── notes.md
├── session-02-3d-features-modeling/    <- Extrude/revolve/fillet → part modeling (2h)
│   └── notes.md
├── session-03-assembly-constraints/    <- Joints, assemblies, motion (2h)
│   └── notes.md
├── session-04-parametric-tolerancing/  <- Parameters, fit/tolerance for 3D printing (2h)
│   └── notes.md
├── session-05-capstone-part-export/    <- Small enclosure, export to STL (2h)
│   └── notes.md
└── progress-tracker.md
```

How to use

1. Install **Fusion 360** (Personal Use license is free) — do this before Session 1.
2. Work session by session; each `notes.md` has a concept summary + a hands-on task at the end.
3. Session 5 is a mini capstone: design a small sensor enclosure (housing) sized to fit an Arduino Uno or Nano — this

becomes the input to the `3d-printing-prep/` pack.

4. Tick off `progress-tracker.md` as you finish each session.

What “10 hours” means here

5 sessions $\times$ 2 hours = 10 hours, matching the table’s 수업시수 exactly. Split each session into two 1-hour blocks (concept, then hands-on) if that fits your schedule better.